

Buy the rumors and sell the news - a sentiment analysis

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Abstract

Financial markets are broadly influenced by information conveyed across all media platforms. Previous research suggests that sentiment extracted from financial news does affect the stock market (Tetlock, 2007), but to what extent does firm-specific sentiment affect a large-cap company that is highly integrated with broad market movements? This study aims to answer and examine the relationship between sentiment, news volume and stock price volatility using Apple Inc. as a case study.

Weekly aggregated financial news data from 2021 to 2024 is quantified through FinBERT to extract sentiment scores. While combining the sentiment score with stock price data and different range-based volatility measures, four regression models are estimated. The models estimated incorporate the following variables: sentiment, news volume, non-linear transformations, interaction variables, trading volume and wide market volatility.

Across the estimated models a clear pattern is discovered. Broad market conditions are the dominant driver for all models, more specifically, wide market volatility and trading volume. However, news volume shows a clear U-shaped non-linear relationship with volatility. Sentiment shows limited explanatory power, but when transforming sentiment into an interaction variable with news volume, sentiment contributes significantly in explaining volatility across all models.

The Johnson-Neyman analysis indicates that the effect of sentiment on volatility is significant during periods of either unusually high news volume or unusually low news volume. This analysis remains consistent across all model specifications.

Overall, the results obtained in this study suggest that firm-specific sentiment only has a marginal effect on the volatility of the particular stock, while the market-wide conditions remain as the dominant driver.

1 Introduction

Financial markets are fundamentally driven by information. Information, conveyed through corporate reports, macroeconomic announcements and financially firm-related news is incorporated into asset prices. While classic financial theory assumes that all information available reflects the price instantly, empirical evidence suggests that interpretation of news and its intensity play an important role in short-term market dynamics.

During the last recent years, the rapid growth of digital financial media has intensified the flow of financially related news to the market participants. This progress has not only increased the interest in what information that is released, but also how the information is conveyed and framed. Sentiment analysis, also called opinion mining, provides a systematic framework for how to quantify the emotional tone in the contextual data. By transforming unstructured financial text into a numerical sentiment-score, researchers are given the opportunity to study how market-sentiment interacts with stock price movements and volatility.

Previous research suggests that sentiment extracted from financial news can influence market movements, especially during times of uncertainty or increased publicity. However, at the same time it is still unclear to what extent the news sentiment meaningfully contributes to explaining the volatility. Large and well-established firms are often integrated with broad market movements which raises the question if firm-related sentiment really impacts the volatility.

Furthermore, our purpose with this study is to examine the relationship between sentiment, news volume and stock market volatility using the company Apple Inc. An extensively media covered company like Apple is a highly suitable subject for analyzing the sentiment effect. This study will not evaluate the sentiment model FinBERT itself but serve as a tool in quantifying the contextual financial tone in news articles related to Apple.

To ensure robustness throughout this study, four different volatility estimators will be used, each capturing different aspects of volatility measures. Through weekly regression models we estimate whether sentiment is directly affecting volatility or if certain transformations such as polynomial, logarithmic or interaction variables are more suitable for capturing and explaining the variation in volatility while controlling for broad market volatility.

Finally, by comparing four different volatility measures and model specification, the main purpose of this study is to examine the effect of sentiment in a large-cap context, rather than frame it as a dominant driver in explaining volatility. This study highlights the conditional nature of sentiment alongside strong systematic market forces

2 Limitations

2.1 FinBERT

It is important to emphasize that the purpose of this study is not to evaluate FinBERT as a Large Language Model. This study only utilizes FinBERT as a way of quantifying sentiment from textual data in order to analyze the economic relationship between sentiment, news volume and stock volatility.

3 Theory

Sentiment analysis, which is also known as opinion mining, is a subfield of Natural Language Processing (NLP). NLP focuses on enabling systems to process and understand human languages. Through algorithms, large unstructured textual data is systematically derived to find valuable insights. As a subfield of NLP, sentiment analysis primary goal is to determine the emotional tone and expressed intensity in textual data. When sentiment is concluded through specific methods, it is often categorized into positive, negative or neutral. Sentiment analysis has broad market applications, such as market analysis, social media monitoring and political forecasting (Zhang, 2024).

Sentiment is not only expressed through individual words, but by context and interpersonal structures which often makes the sentiment analysis complex and highly dependent on deeper understanding. For a model to determine the exact sentiment communicated, it must understand more than just positive or negative loaded words, they must understand how language is used in context to express sentiment. (Zhang, 2024) highlights the difficulties in how the text is communicated, as it is often communicated in subtle ways through certain linguistic structures.

Through the development of Large Language Models (LLM) models, more advanced and complex systems, such as transformer-based models have been introduced to sentiment analysis. In 2019 (Devlin et al., 2019) introduced BERT, a deep bidirectional transformer pretrained on large corporation-related textual data using masked language modelling (MLM) and next sentence prediction (NSP). MLM allows BERT to learn the context of word representations through predictions of masked tokens using both left and right contextual information. With the bidirectional approach, the model is allowed to interpret sentiment through syntactic nuances that serve an essential role for highly accurate sentiment classification.

Previous research has shown that finBERT, an LLM model developed for financial text, consequentially exceeds traditional methods and modern machine and deep learning models for sentiment classification. Compared to algorithms like Naïve Bayes, SVM, Random Forest, CNN and LSTM, finBERT has achieved significantly higher accuracy. According to (Huang et al., 2022), FinBERT has an ability to distinguish between positive or negative sentiment where other models label as “neutral”. The study also shows that finBERT outperforms

lexicon-based models such as Loughran-McDonald lexicon and VADER where the lexicon-based model exhibits a considerably lower accuracy rate in financial contexts compared to finBERT.

3.1 FinBERT

This study utilizes FinBERT, a variant of Bidirectional Encoder Representations from Transformer (BERT). BERT is a Large Language Model (LLM) created by Google in 2019, designed for deep learning application in natural language processing (NLP). FinBERT is an adaption of that model, specifically customized for financially related text. FinBERT builds upon BERT’s and uses the same strategy in leveraging the bidirectional learning strategy, meaning that the finBERT model captures both the preceding and following word in a sentence in order to understand the full contextual meaning (Devlin et al., 2019).

The training process behind BERT and finBERT consists of a two steps: unsupervised pre-training followed by supervised fine-tuning. Unsupervised pre-training includes optimization of the model in Masked Language modelling (MLM) and Next Sentence Prediction (NSP) (Huang et al., 2022)

During the MLM phase, approximately 15 percent of the tokens are randomly masked within each sequence, where the model then is tasked to predict the masked words based on the context of the unmasked tokens in the sequence (Huang et al., 2022). This approach is a central component in the mechanism of BERT’s bidirectionality and forces the model to infer the missing words by applying it. Through this process of predicting the masked tokens, the model learns semantic and syntactic patterns allowing BERT to capture deep linguistic structures. (Jawahar et al., 2019)

In the second component of unsupervised pre-training, NSP, the model is trained on pairwise sentences to determine if the second sentence is truly followed by the first, or if the first sentence has been randomly selected from another paragraph. This mechanism teaches the model in recognition ability and relationships between sentences which is crucial in tasks such as sentiment analysis.

After the pre training, the model undergoes fine-tuning, where it is adapted to specific NLP tasks. Fine-tuning only requires minor adjustments to the parameters making BERT-based models highly efficient and accurate, even on small labeled dataset (Huang et al., 2022).

FinBERT has been pretrained on nearly 5 billion tokens of financial texts, including financial analyst reports, corporate annual and quarterly filings and earnings conference transcripts. This has given the model a fundamental domain-specific understanding of financial language. Through this process, FinBERT exhibits specialized knowledge which allows it to identify and distinguish sentiment where general models often mislabel as neutral. According to (Huang et al., 2022) empirical evaluations of FinBERT, it is shown to significantly outperform traditional machine- and deep learning methods, such as, Naive Bayes, SVM, Random Forest, CNNs and LSTMs, as well as lexicon-based models.

3.2 Ordinary least squares

Ordinary Least Squares (OLS) is the most widely used method in econometrics when quantifying linear relationships between two or more variables. In Basic Econometrics 2009, Gujarati explains that the parameters through OLS, estimate the regression function by fitting a simple regression line that minimizes the distance between the observed value and the line itself (Gujarati and Porter, 2009). To do so, OLS estimates the squared deviations between the actual value, Y_i and the predicted value $\hat{Y}_i$. Through this process, the minimization principle forms the core of the “Least Squares Method” (Gujarati and Porter, 2009)

Historically, the OLS method was created by Carl Friedrich Gauss and Adrien-Marie Legendre in the early 19th century. Gujarati highlights the fact that the method is attributed to Carl Friedrich Gauss, although Legendre was the first ever to publish it in 1805. The reason for this is that Gauss formalized the theoretical properties for it in 1809. However, their findings established the mathematical foundation for estimating the linear relationships through minimization of residuals.

$$\sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

The least squares method ensures that larger deviations receive a disproportionately greater weight through the sum of squared residuals, rather than the sum of raw residuals. This approach avoids situations where positive and negative residuals cancel each other out, despite being far from each other (Gujarati and Porter, 2009). This process provides the optimal estimator for the regression coefficients, given the observed data.

In empirical research, OLS has become a foundational tool due to its mathematical simplicity and strong statistical properties. Under the Gauss-Markov assumptions of the classical linear regression model, the estimates are linear, unbiased and exhibit minimum variance among all linear unbiased estimators. By following the criteria formalized by Gauss-Markov, OLS ensures the Best Linear Unbiased Estimator (BLUE) (Gujarati and Porter, 2009)

In the context of this study, we believe OLS is an appropriate tool to examine whether linear relationships exist between sentiment measures and financial volatility. Since OLS is designed to estimate this type of association and provide an empirical result if sentiment has an effect on volatility, this will be the methodological tool for the study that follows.

3.3 The Gauss-Markov Theorem and BLUE

The Gauss-Markov theorem states the fundamental statistical properties of the OLS estimator. According to (Gujarati and Porter, 2009) the theorem, if fulfilled under the classical linear regression assumptions is the Best Linear Unbiased Estimator (BLUE). The meaning behind it, is that among all estimators that are linear functions of a dependent variable, OLS has the lowest variance while remaining unbiased (Gujarati and Porter, 2009). 2.4.3 The Gauss-Markov assumptions In Basic Econometrics 2009, Gujarati refers to a set of assumptions,

namely the Gauss-Markov assumptions. These assumptions must hold for the OLS to be unbiased and when these are met, the OLS yields BLUE estimators of the regression coefficients (Gujarati and Porter, 2009). These Gauss-Markov assumptions are stated below.

Linearity

The model must be linear and expressed as:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_k X_{ik} + \varepsilon_i$$

The linearity assumptions refer to coefficients and not mainly to the variables. By applying appropriate transformations such as, logarithmic, polynomials or square root, nonlinear relationships can be estimated through OLS. (Gujarati and Porter, 2009)

Independence of errors

$$\text{Cov}(\varepsilon_i, \varepsilon_j) = 0 \quad \text{for } i \neq j$$

Statistical independence refers to that every observation being drawn randomly from the population and that the error term being must be independent across all observations. If the errors appear to be dependent of each other, which often occurs in time series data, the OLS estimator may no longer be unbiased. To account for dependent observations, lagged variables can be included to remedy the violation. (Flatt and Jacobs, 2019)

Homoskedasticity

The variance of the error term must be consistent across all observations of all predicted variables:

$$\text{Var}(\varepsilon_i) = \sigma^2$$

If this assumption fails to be fulfilled, the OLS remains unbiased, but the standard errors are no longer reliable. (Flatt and Jacobs, 2019)

No perfect multicollinearity

$$X_j \neq a_1 X_1 + a_2 X_2 + \cdots + a_{j-1} X_{j-1}$$

No perfect multicollinearity can be present, meaning that no independent variable is allowed to perfectly correlate with another independent variable. Perfect multicollinearity prevents calculations of the OLS estimates. Even though perfect multicollinearity is a rare phenomenon, high correlations still inflate variances of the estimated coefficients, making interpretability more complicated. (Flatt and Jacobs, 2019)

Normality

$$\varepsilon_i \sim N(0, \sigma^2)$$

Normality itself is not required for the OLS to be unbiased or consistent, but Gujarati. Basic Economics 2009, highlights the importance of normally distributed error terms, which are important and allow for valid t-tests, F-tests and constructions of confidence intervals (Gujarati and Porter, 2009). Normality of errors is therefore important in hypothesis testing and even more important for smaller datasets (Flatt and Jacobs, 2019).

3.4 Volatility estimation

Volatility is a measurement of risk of asset return and is shown by the standard deviations of the returns for a given period of time.

As true volatility cannot be measured (Ahoniemi and Lanne, 2013) it has to be estimated by using data. A simple form of estimation of realized volatility from daily closing prices, which is widely used in the literature (Miralles-Quirós et al., 2019), is the mean of the squared daily returns. Realized volatility can be formally defined as

$$RV = \sum_{i=1}^n r_i^2$$

where $r_i = \ln(\frac{C_i}{C_{i-1}})$, and C_i is the closing price of an asset on day i over the period of n days.

This measurement can however be noisy (Miralles-Quirós et al., 2019) and does not take into account for any intraday fluctuations that occur during the day. If the closing price of an asset is similar or equal, the actual volatility may not be accurately calculated as the squared daily return would suggest identical volatility. To more accurately estimate the true volatility, range-based estimators are used. Range-based estimators are effectively used when only daily data is available in the form of daily opening, high, low and closing prices, furthermore denoted as O , H , L and C . These estimators can capture the intraday volatility by including H and L .

The first range-based volatility estimation method, Parkinson's PK measure (Khoo et al., 2025) measure which uses the daily H and L . It is formally defined as

$$\sigma_{PK}^2 = \frac{1}{4\ln 2} [\ln(\frac{H_t}{L_t})]^2.$$

A further extension of Parkinson's PK measure is the Garman and Klass volatility estimate, which includes opening and closing prices.

$$\sigma_{GK}^2 = 0.511 [\ln(\frac{H_t}{L_t})]^2 - 0.386 [\ln(\frac{C_t}{O_t})]^2.$$

The PK measure as well as Garman and Klass volatility estimate works under the assumption that there is no drift present (Rogers et al., 1994). If drift is

present, such as a general positive or negative trend, Rogers and Satchell estimation of volatility remains unbiased and outperforms the Garman-Klass estimation according to (Rogers et al., 1994). It is formally defined as

$$\sigma_{RS}^2 = \ln\left(\frac{H_t}{O_t}\right)\left[\ln\left(\frac{H_t}{O_t}\right) - \ln\left(\frac{C_t}{O_t}\right)\right] + \ln\left(\frac{L_t}{O_t}\right)\left[\ln\left(\frac{L_t}{O_t}\right) - \ln\left(\frac{C_t}{O_t}\right)\right].$$

One downside to this measurement is that this volatility estimation can have terms that become 0, if for example $H = C$ or $L = O$ as shown by (Miralles-Quirós et al., 2019). This can lead to the whole expression giving zero volatility which makes the estimation unreliable.

The last volatility measurement that is used is presented by (Yang and Zhang, 2000). It is formally defined as

$$\sigma_{YZ}^2 = \sigma_o^2 + k\sigma_c^2 + (1 - k)\sigma_{RS}^2 \text{ where}$$

$$\sigma_o^2 = \ln\left(\frac{O_t}{C_{t-1}}\right),$$

$$\sigma_c^2 = \ln\left(\frac{C_t}{O_t}\right),$$

$$k = \frac{\alpha - 1}{\alpha + \frac{n+1}{n-1}}.$$

Yang-Zhang decomposes daily volatility into an overnight component and an intraday component. By averaging these components, the Yang-Zhang estimator captures the total daily variance rather than isolating short-term intraday fluctuations. In relation to close-to-close estimators such as Parkinson's or Garman-Klass, the Yang-Zhang volatility is independent of both the drift and opening jumps of the underlying price movement, making the Yang-Zhang volatility the estimator with the lowest theoretical variance (Yang and Zhang, 2000).

4 Data

The data used in this thesis consist of financial market data and news-based textual data spanning the period from 2016 to 2024. As the news data do not include exact publication timestamps, it is not possible to determine the precise timing of when individual articles became available to market participants during a trading day. Therefore, news articles cannot be reliably matched to their corresponding trading day. To address this issue, news articles are aggregated to a weekly frequency and merged with weekly financial data. Weeks are defined as running from Saturday to Friday, ensuring that weekend news is incorporated into the subsequent trading week, with all articles assigned to weeks based on their publication dates.

4.1 News articles

The news articles that are used in our analysis are obtained from the Kaggle dataset *Apple Stock (AAPL): Historical Financial News Data*. The dataset consists of a 29.752 entries of several variables, such as publication date, article title, text content, URL link, and metadata related to stock symbols, tags, and sentiment scores. For this study we only use the URL links to news articles related to Apple Inc., covering the period from 2016 to 2024 (Ossai, 2024).

Through a Python script, we then run a loop using the article scraping package `newspaper3k` to download the following variables:

1. `url`: HTML-link to the specific article
2. `title`: Title of the news article
3. `text`: The body text of the news article
4. `date`: The publishing date of the article

These variables are saved in to a `pandas` data frame, containing 23.415 articles. Approximately 6000 articles are lost as the script is not able to extract the news article. As our analysis is focused on Apple specific company news we remove news not mentioning the `AAPL` ticker name in the `text` variable. This filtering step helps exclude macroeconomic or broad market commentary that may have been tagged with `AAPL` in the original dataset but does not contain firm-specific information. Further, the dates for a large portion of the new `pandas` data frame are not successfully extracted. To resolve this, we match the corresponding dates and URLs from the original Kaggle dataset with our new `pandas` data frame.

4.2 Financial data

Financial data for `AAPL` stock is obtained through the `tidyquant` package in R. The dataset contains of 2735 observations ranging from 2015 to 2025 with the following variables

1. `symbol`: Ticker symbol
2. `date`: Date, format YYYY-MM-DD
3. `open`: Opening price of the stock
4. `high`: Highest price during the trading day
5. `low`: Lowest price during the trading day
6. `close`: Closing price of the trading day
7. `volume`: Volume of the trading day

8. **adjusted**: Closing price of the trading day, adjusted for dividends/split

Through the variables **open**, **high**, **low** and **close** (OHLC) we calculate the range-based volatility estimations we use for our analysis. We then match the corresponding trading day from the **date** with the news article and sentiment.

Further, financial data for S&P500 is obtained through the **tidyquant** package in R. The dataset contains of 2735 observations ranging from 2015 to 2025 with the following variables

1. **symbol**: Ticker symbol
2. **date**: Date, format YYYY-MM-DD
3. **open**: Opening price of the stock
4. **high**: Highest price during the trading day
5. **low**: Lowest price during the trading day
6. **close**: Closing price of the trading day
7. **volume**: Volume of the trading day
8. **adjusted**: Closing price of the trading day, adjusted for dividends/split

Through the variables **open**, **high**, **low** and **close** (OHLC) we calculate the range-based volatility estimations we use for our analysis. We then match the corresponding trading day from the **date** with the news article and sentiment.

5 Method / Model

5.1 Volatility calculations

To calculate the weekly volatility we use daily data and average it over the week where $n = 4$ or $n = 5$ depending on how many days the market is open for a given week. This is done to get comparable estimations where the time periods differ (Corsi, 2009). The volatilities are calculated for **AAPL** and **S&P500**.

For Parkinson's PK measure we calculate

$$\sigma_{PK} = \sqrt{\frac{1}{n} \sum_{i=1}^n \left(\frac{1}{4 \ln 2} [\ln(\frac{H_t}{L_t})]^2 \right)}$$

$$\sigma_{PK_{adj}} = \sqrt{\frac{1}{n} \sum_{i=1}^n \left(\frac{1}{4 \ln 2} [\ln(\frac{H_t}{L_t})]^2 + j^2 \right)}$$

Further, the Garman Klass volatility is calculated as

$$\sigma_{GK} = \sqrt{\frac{1}{n} \sum_{i=1}^n (0.511 [\ln(\frac{H_t}{L_t})]^2 - 0.386 [\ln(\frac{C_t}{O_t})]^2)}$$

The Rogers Satchell volatility is calculated as

$$\sigma_{RS} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\ln(\frac{H_t}{O_t})[\ln(\frac{H_t}{O_t}) - \ln(\frac{C_t}{O_t})] + \ln(\frac{L_t}{O_t})[\ln(\frac{L_t}{O_t}) - \ln(\frac{C_t}{O_t})])}$$

Lastly, the Yang Zhang drift independent volatility is calculated as

$$\sigma_{YZ} = \sqrt{\frac{1}{n} \sum_{i=1}^n \sigma_o^2 + k\sigma_c^2 + (1 - k)\sigma_{RS}^2}$$

Where $\alpha = 1.34$ is used to calculate k as suggested by (Yang and Zhang, 2000) for optimal calibration of k .

5.2 Data handling

After extracting the sentiment score for the individual news, the data are merged by date and we remove any duplicate entries based on article titles.

As time stamps of the news articles are not available, the exact timing of when the published article actually affects the market is uncertain. To reduce this uncertainty we assume the news published on Saturday and Sunday affects the volatility on the following Monday.

To construct a weekly structure, each date is assigned to a corresponding week start, where we define a week beginning on Saturday and ending on Friday. The total trading volume for each week is obtained by summing daily volumes, and the sentiment score for each week is obtained by summing the weekly sentiment score. Lastly, we add a dummy variable with the value 1 if a quarterly report is released that day, 0 otherwise. The weeks aggregate the daily data as

- **open:** Opening price on Monday
- **high:** Maximum price of the week
- **low:** Minimum price of the week
- **close:** Closing price on Friday
- **volume:** Volume of the trading week
- **sentiment_sum:** Weekly sum of sentiment score
- **news_count:** Weekly sum of news articles
- **report:** Dummy variable if quarterly report is released that day

The daily volatilities are then averaged and assigned to their corresponding week to ensure alignment between the news variables and weekly volatility measures.

Finally, as weekly news data is only sequentially available for the period January 2021 to November 2024, all observations preceding or following this period are removed.

5.3 Linear regression models

To reduce multicollinearity between the `sentiment_sum` and `news_count` variables, both variables are mean-centered, meaning the sample mean is subtracted so that each variable has a mean of 0. This transformation is particularly important when interactions are included in regression models, as centering reduces correlations among variables (Shieh, 2011).

These new variables are denoted with a subscript c . This gives us our base model formulated as

$$\begin{aligned}\log(\text{Volatility}_t) = & \beta_0 + \beta_1 \log(\text{Volatility}_{t-1}) \\ & + \beta_2 (\text{Sentiment}_{c,t} \cdot \text{news}_{c,t}) \\ & + \beta_3 \text{News}_{c,t} + \beta_4 \text{Sentiment}_{c,t} \\ & + \beta_7 \log(\text{Volume}_t) \\ & + \beta_8 \log(\text{Volatility}_{S\&P500,t}) \\ & + \beta_9 \log(\text{Volatility}_{S\&P500,t-1}) \\ & + \beta_{10} \text{Report}_t + \varepsilon_t\end{aligned}$$

Further, as the financial data are typically heavy tailed or right skewed (Eom et al., 2019) we apply the natural logarithm to the volatility estimators as well as to the trading volume variable. This transformation reduces skewness and stabilizes variance, improving the suitability of the data for linear regression analysis.

To evaluate the linear regression models, combined with visual inspection of the data, we apply Ramsey’s RESET test for functional form misspecification (Ramsey, 1969). The test examines whether nonlinear transformations of the fitted values provide additional explanatory power beyond the specified model. Further, we use Breusch-Pagan test for homoscedasticity (Breusch and Pagan, 1979), Box-Ljung for autocorrelation (Box and Ljung, 1978), VIF (Variance inflation factor) for multicollinearity and Shapiro-Wilks test for normality of error terms (Shapiro and Wilk, 1965).

5.4 Interaction

To evaluate the interaction variable we use interaction plots using the `emmip` plot (Estimated Marginal Means Interaction Plot) from the `emmeans` package in R and Johnson-Neyman plots. The `emmip` plot displays the predicted values of the outcome variable for different levels of sentiment separately for three chosen levels of news articles published. As these variables are centered, we chose three levels of news articles published of low, medium and high. A low level of news articles published is defined as one standard deviation under the mean, a medium level of news articles published is defined as the mean and lastly, a high level of news articles published is defined as one standard deviation over the mean.

The Johnson-Neyman plot (Lin, 2020) illustrates how the relationship between sentiment and the volatility estimate changes depending on the amount of news articles published where it displays the marginal effect of sentiment on the given volatility estimate. Further, the Johnson-Neyman plot displays confidence intervals which identify specific regions in which the effect is significant where the confidence interval does not include zero.

5.4.1 Parkinson

A preliminary RESET test for the original model using Parkinson volatility indicated that the functional form misspecification. To remedy this, we apply second-order polynomial terms to the variables `news_c` and `sentiment_c`. The inclusion of these polynomial terms successfully eliminated the model misspecification issue and significantly improved the model fit, making it suitable for analysis. This gives our Parkinson volatility estimate model specified as

$$\begin{aligned}\log(\text{Parkinson}_t) = & \beta_0 + \beta_1 \log(\text{Parkinson}_{\text{Apple}_{t-1}}) \\ & + \beta_2(\text{Sentiment}_{c,t} \cdot \text{News}_{c,t}) \\ & + \beta_3\text{News}_{c,t} + \beta_4\text{Sentiment}_{c,t} \\ & + \beta_5\text{News}_{c,t}^2 + \beta_6\text{Sentiment}_{c,t}^2 \\ & + \beta_7 \log(\text{Volume}_t^2) \\ & + \beta_8 \log(\text{Parkinson}_{\text{S\&P500}_t}) \\ & + \beta_9 \log(\text{Parkinson}_{\text{S\&P500}_{t-1}}) \\ & + \beta_{10}\text{Report}_t + \varepsilon_t\end{aligned}$$

5.4.2 Garman Klass

Similar to the Parkinson model, applying the original model to the Garman Klass volatility estimate, the model exhibited functional form misspecification according to the RESET test. The same solution is applied to the Garman Klass model where the variables `sentiment_c` and `news_c` are squared and added to the linear regression model. This gives our model specification for the Garman Klass volatility as

$$\begin{aligned}
\log(\text{GarmanKlass}_{\text{Apple}_t}) = & \beta_0 + \beta_1 \log(\text{GarmanKlass}_{\text{Apple}_{t-1}}) \\
& + \beta_2(\text{Sentiment}_{c,t} \cdot \text{News}_{c,t}) \\
& + \beta_3 \text{News}_{c,t} + \beta_4 \text{Sentiment}_{c,t} \\
& + \beta_5 \text{News}_{c,t}^2 + \beta_6 \text{Sentiment}_{c,t}^2 \\
& + \beta_7 \log(\text{Volume}_t^2) \\
& + \beta_8 \log(\text{GarmanKlass}_{\text{S\&P500}_t}) \\
& + \beta_9 \log(\text{GarmanKlass}_{\text{S\&P500}_{t-1}}) \\
& + \beta_{10} \text{Report}_t + \varepsilon_t
\end{aligned}$$

5.4.3 Rogers Satchell

The Rogers Satchell specification also exhibited functional-form misspecification without polynomial terms. As with the previous two models, adding second-order polynomial variables of sentiment_c and news_c resolved these issues and the polynomial structure is retained for the Rogers Satchell model as well. This gives our modified model for Rogers Satchell volatility estimation as

$$\begin{aligned}
\log(\text{RogersSatchell}_{\text{Apple}_t}) = & \beta_0 + \beta_1 \log(\text{RogersSatchell}_{\text{Apple}_{t-1}}) \\
& + \beta_2(\text{Sentiment}_{c,t} \cdot \text{News}_{c,t}) \\
& + \beta_3 \text{News}_{c,t} + \beta_4 \text{Sentiment}_{c,t} \\
& + \beta_5 \text{News}_{c,t}^2 + \beta_6 \text{Sentiment}_{c,t}^2 \\
& + \beta_7 \log(\text{Volume}_t^2) \\
& + \beta_8 \log(\text{RogersSatchell}_{\text{S\&P500}_t}) \\
& + \beta_9 \log(\text{RogersSatchell}_{\text{S\&P500}_{t-1}}) \\
& + \beta_{10} \text{Report}_t + \varepsilon_t
\end{aligned}$$

5.4.4 Yang Zhang

While the previous models exhibit functional-form misspecification without polynomial terms, the Yang Zhang model does. Although, added polynomial terms improve the models performance which is why we include it. This model exhibits autocorrelation in its error terms which is why a lagged variable for the volatility estimate of S&P500 is added to address this issue, leaving the modified model specification as follows.

$$\begin{aligned}
\log(\text{YangZhang}_{\text{Apple}_t}) = & \beta_0 + \beta_1 \log(\text{YangZhang}_{\text{Apple}_{t-1}}) \\
& + \beta_2(\text{Sentiment}_{c,t} \cdot \text{News}_{c,t}) \\
& + \beta_3 \text{Sentiment}_{c,t} + \beta_4 \text{News}_{c,t} \\
& + \beta_5 \text{Sentiment}_t^2 + \beta_6 \text{News}_t^2 \\
& + \beta_7 \log(\text{Volume}_t^2) \\
& + \beta_8 \log(\text{YangZhang}_{\text{S\&P500}_t}) \\
& + \beta_9 \log(\text{YangZhang}_{\text{S\&P500}_{t-1}}) \\
& + \beta_{10} \text{Report}_t + \varepsilon_t
\end{aligned}$$

6 Result

6.1 Model summary and main findings

This study estimates four linear regression models using four different volatility measures: Parkinson, Garman-Klass, Roger-Satchell and Yang-Zhang. They all aim to explain the weekly variation in volatility using primarily sentiment and news volume, but also an interaction variable of the two, trading volume and S&P500 volatility. Across the four model specifications, a number of consistent empirical patterns are recurrent:

6.2 Dominance of broad market conditions

Throughout all four models, the S&P500 volatility is the strongest predictor of volatility, regardless of the estimator used. Estimated coefficient values range from 0.42 to 0.51 for every model specification, confirming that broad market conditions is the most dominating factor.

This means that Apple's volatility, regardless of the estimator of volatility used behaves as a function of broad market movements, rather than a function of sentiment and news intensity which is theoretically expected from a financial point of view. Baker and Wurgler explain that well established companies tend to follow the broad market sentiment, namely interest rates and overall index movements. This means that specific company-related news article has a smaller impact on returns and volatility compared to the influence of the wider market (Baker and Wurgler, 2006).

6.3 Non-linear impact of news volume

In all models the influence of news volume is noticeably non-linear given the functional-form test. For all four models, the regression yields a U-shaped relationship, meaning that the non-quadratic news variable is significantly negative while the quadratic news variable is significantly positive, which is further confirmed by the emmip plots shown in the appendix. This indicates that when

news volume increases from a low to moderate level, it tends to dampen the volatility and when news volume starts to increase to a high level, the volatility increases with it.

6.4 Sentiment effects: weak in isolation, meaningful in interaction

Across all estimated models, the untransformed linear sentiment variable remains non-significant which means sentiment alone is not able to strongly predict the volatility. However, by transforming it into a quadratic variable and by constructing an interaction variable of news count and sentiment, it contributes significantly to explaining the volatility across all models.

In all models the quadratic sentiment variable is highly significant. The positive coefficient indicates that more extreme sentiment, regardless if positive or negative, is associated with increased volatility. In other words, it is either low or high levels of sentiment that is giving the effect. For the quadratic news variable, more news tends to influence the increase in volatility.

Furthermore, the interaction effect between sentiment and news volume is included in all models. The sentiment-news interaction is significant and has a negative coefficient. The negative coefficient implies that the marginal effect of sentiment on volatility decreases as news volume increases. This indicates that when there are few articles released during the week, the sentiment has a stronger effect on volatility compared to when more articles are released.

These findings align with the Johnson-Neyman plot for all models which indicates that sentiment has a significant effect primarily when either news volume is very high or very low, but not when news volume is in a "moderate" range. Overall, the Johnson-Neyman plot indicates that the effect of sentiment is primarily driven by the amount of news volume.

6.5 Model assumptions and Robustness Across Volatility Measures

Across the estimated models, diagnostic tests generally support the sufficiency of all regression models. Ramsey's RESET test indicates that no model misspecification has been done, meaning that all models follow the correct functional form. The Breusch-Pagan test suggests that almost all models support the assumption of homoskedasticity, although the Yang-Zhang model shows some evidence of heteroskedastic residuals. The Box-Ljung test indicates autocorrelation at different levels of lags throughout all model specifications, where the HAC robust standard errors are the same for every lag.

The normality assumption for the error terms is frequently violated. The Parkinson and Garman-Klass show clear deviations from normality in both graphical diagnostics and the Shapiro-Wilks test. These results limit the ability of hypothesis testing, but do not affect the OLS coefficient as it remains unbiased even when the assumption of normality is violated. The Rogers-Satchell model shows residuals closer to normality than both Parkinson and Garman-Klass,

while the residuals of the Yang-Zhang model is almost normally distributed. Overall, inferential results should be interpreted with caution, as some of the models violate the assumptions of normality.

Despite differences in how volatility is estimated and minor variations in model specification, the empirical results remain consistent across all models. Key coefficients remain statistically significant throughout all models and cluster around similar estimates, indicating that the findings are not concluded by the choice of a certain volatility estimator. The explanatory power and adjusted R^2 confirm this, as they also cluster tightly around similar levels.

The result of the diagnostics done suggests that the empirical findings are robust. However, some assumptions, primarily normality, are not perfect, but the overall stability in all tests strengthens the validity of the conclusions drawn for this analysis.

The coefficient estimations with their HAC robust p-values are presented in Table 1 below.

Variable	Parkinson	Garman-Klass	Rogers-Satchell	Yang-Zhang
$Volatility_{Apple,t-1}$	0.1247 (0.1791)	0.1968 (0.0248)	0.1968 (0.0248)	0.1030 (0.1124)
$Sentiment_c$	0.0013 (0.5586)	0.0026 (0.2297)	0.0026 (0.2297)	0.0005 (0.8231)
$News_c$	-0.0108 (0.000)	-0.0093 (0.000)	-0.0093 (0.000)	-0.0094 (0.000)
$News_c^2$	0.0003 (0.000)	0.0002 (0.000)	0.0002 (0.000)	0.0002 (0.0011)
$Sentiment_c^2$	0.0005 (0.000)	0.0004 (0.0002)	0.0004 (0.0002)	0.0005 (0.0293)
$\log(volume^2)$	0.0074 (0.000)	0.0062 (0.000)	0.0062 (0.000)	0.0072 (0.000)
$\log(Volatility_{S\&P500,t})$	0.4271(0.000)	0.4500 (0.000)	0.4500 (0.000)	0.5092 (0.000)
Report	0.0190 (0.7298)	0.0230 (0.6691)	0.0230 (0.6691)	0.1696 (0.0124)
$Sentiment_c * News_c$	-0.0004 (0.0009)	-0.0004 (0.0003)	-0.0004 (0.0003)	-0.0003 (0.1782)
$\log(Volatility_{S\&P500,t-1})$	-0.0496 (0.4589)	-0.0954 (0.1328)	-0.0954 (0.1328)	-0.0735 (0.1889)
$R^2_{Adjusted}$	0.7258	0.7284	0.6903	0.7353

Table 1: Coefficient estimates with HAC robust p-values in parenthesis for different volatility measures

Table 1 presents a summary of all the model specifications. The volatility formulas used are, Parkinson, Garman-Klass, Rogers-Satchell and Yang-Zhang for each dependent variable. What can be seen are clear patterns that emerge across all models.

One of the most prominent findings is the dominant role of broad market conditions. The coefficient of S&P500 is positive and highly significant across all models. The estimate ranges from 0.43 in the Parkinson model to 0.51 in the Yang-Zhang model, indicating the Yang-Zhang is formulated, captures the variation in volatility the most. However, this result shows that a one percent increase in volatility is associated with a substantial increase in the volatility of the Apple.Inc stock. This is also a confirmation of how systemic market risk is the dominant driver of the volatility.

Trading volume exhibits a positive and highly significant coefficient across

all models, with coefficients ranging from 0.006 to 0.007, indicating the close relationship between trading activity and volatility.

As the table shows, news related variables suggest a non-linear relationship where the linear variable is significantly negative, ranging from -0.009 to -0.011, while the quadratic variable is positive and ranges from around 0.0002 to 0.0003, while remaining significant. This finding suggests a U-shaped relationships where low to moderate news volume tends to dampen the volatility, whereas intensive news coverage is associated with increased volatility.

The variables regarding sentiment show a more nuanced pattern. The linear sentiment variable remains insignificant across all model specifications, suggesting that the isolated effect of sentiment does not explain any variation of volatility. However, the quadratic sentiment coefficient is positive and significant across all models, ranging from 0.0004 to 0.0005, indicating that extreme sentiment regardless of the sign, is associated with higher volatility. The interaction variable between sentiment and news volume is negative and significant for all models, except Yang-Zhang. The coefficient is estimated around -0.0004 which suggests that the marginal effect of of sentiment decreases as news volume increases. As this interaction variable is not significant for the Yang-Zhang specification, it is clear that the interaction variable is sensitive to the choice of volatility estimator.

Overall, as concluded in the paragraph of the main findings, it is clear that market-wide volatility and trading volume are dominating the explanation of variation in the volatility of Apple, while sentiment and news variables contributes conditional through interaction and non-linear effects.

6.6 Joint hypothesis test

To further test the effect of sentiment on the estimated volatilities, two Wald tests for joint significance are performed. The test compares a whether specified coefficients are equal to zero against a full model.

For readability we name the two tests A and B, where test A only test coefficients for variables containing the sentiment variable and test B test coefficients for variables containing the news variables in addition to coefficients for variables containing sentiment. This gives the null and alternative hypothesis as

$$\begin{aligned} H_{0,A} : & \beta_{Sentiment_c} = \beta_{Sentiment_c^2} = \beta_{Sentiment_c * News_c} = 0 \\ H_{1,A} : & \text{At least one of } \beta_{Sentiment_c}, \beta_{Sentiment_c^2}, \beta_{Sentiment_c * News_c} \neq 0 \end{aligned}$$

and

$$\begin{aligned} H_{0,B} : & \beta_{Sentiment_c} = \beta_{Sentiment_c^2} = \beta_{Sentiment_c * News_c} = \beta_{News_c} = \beta_{News_c^2} = 0 \\ H_{1,B} : & \text{At least one of } \beta_{Sentiment_c}, \beta_{Sentiment_c^2}, \beta_{Sentiment_c * News_c}, \beta_{News_c}, \beta_{News_c^2} \neq 0 \end{aligned}$$

As some models exhibited heteroscedasticity and autocorrelation the inference is based on HAC estimators. This also ensures a robust result for those models that do not exhibit heteroscedasticity. The result is presented in Table 2 below

Volatility measure	Test	F-statistic	P-value
Parkinson	A	7.4555	1e-04
Parkinson	B	10.963	0.000
Garman-Klass	A	6.6181	3e-04
Garman-Klass	B	9.9972	0.000
Rogers-Satchell	A	6.7723	2e-04
Rogers-Satchell	B	9.6629	0.000
Yang-Zhang	A	1.7133	0.1656
Yang-Zhang	B	10.011	0.000

Table 2: Joint hypothesis test results

For the volatility measures for Parkinson, Garman-Klass and Rogers-Satchell, the null hypothesis that sentiment has no effect is rejected. In contrast, for the Yang-Zhang volatility measure the effect of sentiment alone is not rejected. Further, when sentiment and news variables are tested jointly, the null hypothesis is rejected for all volatility measures. This indicates that sentiment influences volatility either directly through intraday trading or indirectly through an interaction with news intensity.

7 Discussion

To investigate the impact of news sentiment on the volatility of AAPL, we estimate weekly regressions using multiple volatility measures. As functional-form tests indicate that the relationship between news, sentiment and volatility is non-linear we add polynomial variables as well as an interaction variable. This captures the relationship between the amount of news articles published and its sentiment score with volatility. The squared variables for news and sentiment reveal significant non-linear effects across all volatility measures, indicating that the impact of sentiment on volatility accelerates for extreme levels of news intensity. These results suggest that volatility does not respond proportionally to neither sentiment or news volume, but rather increases at an increasing rate when either variable is extreme.

Across most measures, the results indicate a statistically significant relationship between news variables, though the impact of news sentiment differs depending on the volatility measure. Specifically the Yang-Zhang volatility measure shows a weaker relationship compared to the other volatility measures. While Parkinson, Garman-Klass and Rogers-Satchell volatility measures primarily focus on intraday movements using OHLC prices, the Yang-Zhang measure additionally accounts for overnight returns. As the volatility measure averages

the intraday component and overnight returns, the intraday volatility is diluted by the overnight component. As a result, any volatility spikes induced by either the amount of published news, sentiment score or any interaction of the variables are possibly diluted by the Yang-Zhang volatility. This can explain why the sentiment alone is not statistically significant while the opposite is true for the other intraday focused volatility measures.

Further, as the timestamps of the news articles are not available it could be that all or the majority of news articles are published during trading hours. The efficient market hypothesis says that markets react instantly to new information (Lehmann, 1990). If the news articles available for this study are predominantly published during trading hours it is priced in immediately, affecting the intraday components and not the overnight components. This may also help explain why sentiment and news variables are less informative for overnight-inclusive volatility measures, where price adjustments occur primarily through discrete opening jumps.

This interpretation is consistent with the joint hypothesis test results, which show that news variables contribute explanatory power when tested jointly with sentiment variables, particularly for volatility measures capturing intraday price variation. It is also supported by the `emmip` and Johnson-Neyman plots. The `emmip` plots consistently show that the effect of sentiment on volatility varies with news volume for intraday volatility measures, while the Johnson-Neyman intervals indicate that the marginal effect of sentiment is statistically significant only within specific ranges of news intensity. For the Yang-Zhang volatility measure these conditional effects are statistically insignificant. This suggests that the sentiment-driven effects primarily affect volatility during continuous trading, i.e not during pre- or post market hours. The overnight volatility reflects information accumulation that is less linked to same-day sentiment.

Overall, the results indicate that news sentiment does have a statistically and economically meaningful impact on volatility, particularly during trading hours. More extreme sentiment, whether it is very positive or very negative is associated with higher volatility shown by the nonlinear relationship. The effect of sentiment is conditional on the level of news intensity, as high sentiment has the strongest impact when news volume is low to moderate while its marginal effect diminishes during periods of intense news coverage.

7.1 Direction for future research

Throughout this study, we rely on weekly aggregate measures of both sentiment and volatility. According to the efficient market hypothesis, financial markets often react immediately to released information and by the use of weekly aggregate information (Lehmann, 1990). By doing so, we miss short-lived volatility responses that often occur within hours or minutes followed by a news publication. We therefore urge future research to study intraday or daily data against news articles with exact timestamps.

The aggregation of news article as weekly also introduces time uncertainty in

this study. Throughout this study, weeks are constructed by defining Saturday as start of the week and ending on Fridays, as Saturdays and Sundays news are priced into the market on Mondays. However, due to the absence of an exact timestamp it is impossible to determine if news published on Fridays were published before open, during open or after open. Therefore, we included all the news published on Fridays into that specific week. Future research should therefore gather timestamped news data as it gives a more accurate alignment between sentiment arrival and instant market response.

Regarding firm selection in this study, we focus on a large-cap and well-established firm. This may cause some issues as it is highly integrated with market-wide dynamics, which is concluded in this study. By being integrated with market-wide dynamics including interest rates, macroeconomic conditions and general index movements, specific firm-related news sentiment may be dampened by the overall market sentiment. By studying a smaller firm, which is less correlated to broad market movements, it could provide additional insights into how volatility is affected by news sentiment and if the effects are even bigger.

Moreover, throughout this research we have adopted an empirical framework that is associative rather than casual. While including lagged variables and interaction variables, we have mitigated some concerns but the models presented do not fully address potential endogeneity between the variables we are interested in studying, particularly, volatility, news volume and sentiment. This means that increased volatility might by itself attract media coverage or influence the news sentiment that are being published. Therefore, we suggest that future research regarding this topic may apply causal identification strategies to isolate the raw impact of the sentiment.

Finally, it is also important to address the limited interpretability in the Large Language Model, FinBERT. While most LLM’s inherit ”black-box” characteristics, it is often difficult or even impossible to determine why or how an LLM model makes some decisions. In this case, we don’t the exact way of how FinBERT classified the sentiment and its way of thinking, only that it yielded statistically meaningful data that contributed to significant results. For similar future research, we suggest a combination of both LLMs and perhaps lexicon-based models in order to enhance transparency and robustness in sentiment classification.

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8 Appendix

Model summaries

Volatility measure	R^2	R^2_{adj}	RESET	Breusch-Pagan	Shapiro-Wilks
Parkinson	0.738	0.7258	0.3149	0.3299	0.001
Garman-Klass	0.7419	0.7284	0.287	0.226	0.0022
Rogers-Satchell	0.7057	0.6903	0.2943	0.1489	0.1662
Yang-Zhang	0.7484	0.7353	0.1989	0.0015	0.0488

Table 3: Model evaluation metrics for the volatility measures. For the tests RESET, Breusch-Pagan and Shapiro-Wilks the p-values are shown

Variable	Parkinson	Garman-Klass	Rogers-Satchell	Yang-Zhang
$Volatility_{Apple,t-1}$	2.71	2.79	2.64	2.51
$Sentiment_c$	4.00	3.97	3.94	3.92
$News_c$	6.73	6.79	6.76	6.58
$News_c^2$	4.15	4.18	4.17	4.17
$Sentiment_c^2$	3.58	3.57	3.56	3.52
$\log(volume^2)$	4.10	4.06	3.96	3.87
$\log(Volatility_{S\&P500,t})$	2.11	2.19	1.98	1.95
Report	1.32	1.31	1.32	1.34
$Sentiment_c * News_c$	7.93	7.95	7.95	7.91
$\log(Volatility_{S\&P500,t-1})$	2.96	3.19	2.93	3.12

Table 4: Variance Inflation Factors (VIF) for all regression models. Values above 5 indicate potential multicollinearity.

8.1 Parkinson model

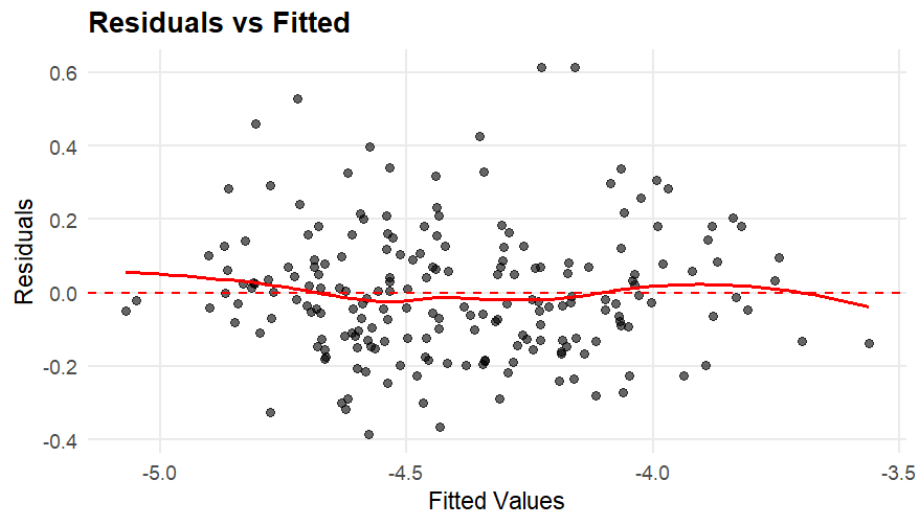


Figure 1: Residual plot for the Parkinson model.

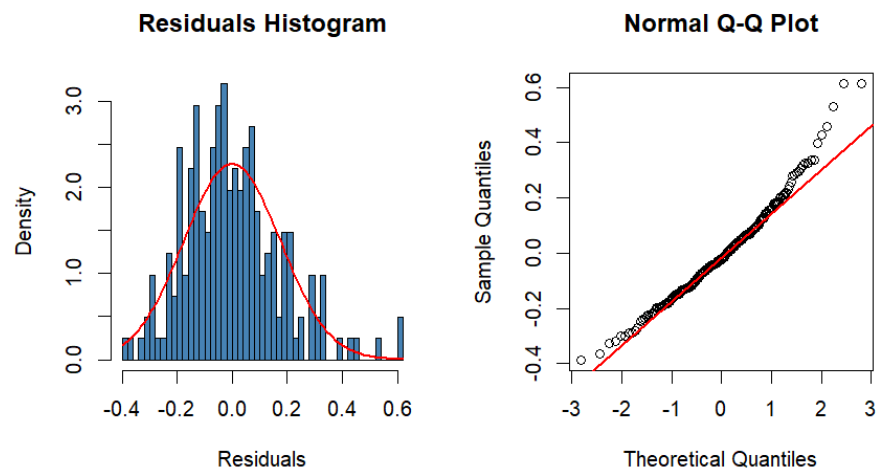


Figure 2: Histogram and QQ plot of residuals for the Parkinson model

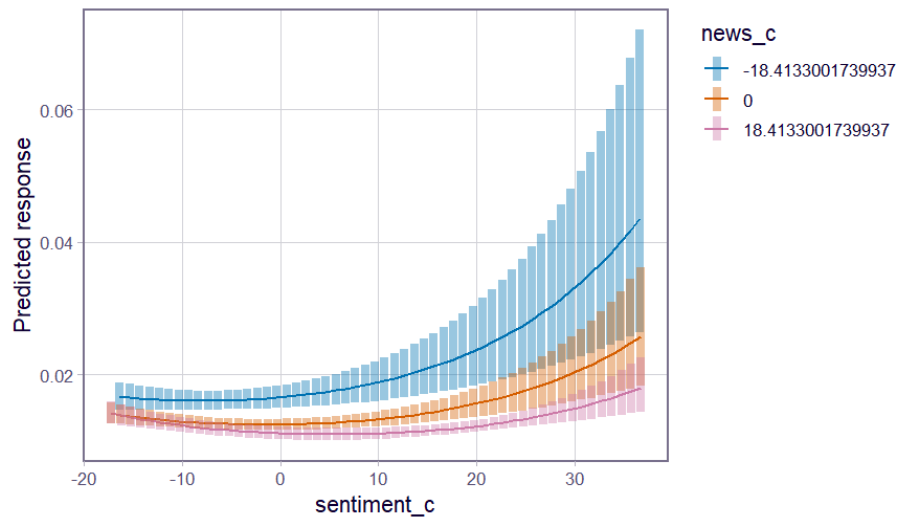


Figure 3: Emmip plot for Parkinson volatility for three levels of news (low, mean and high)

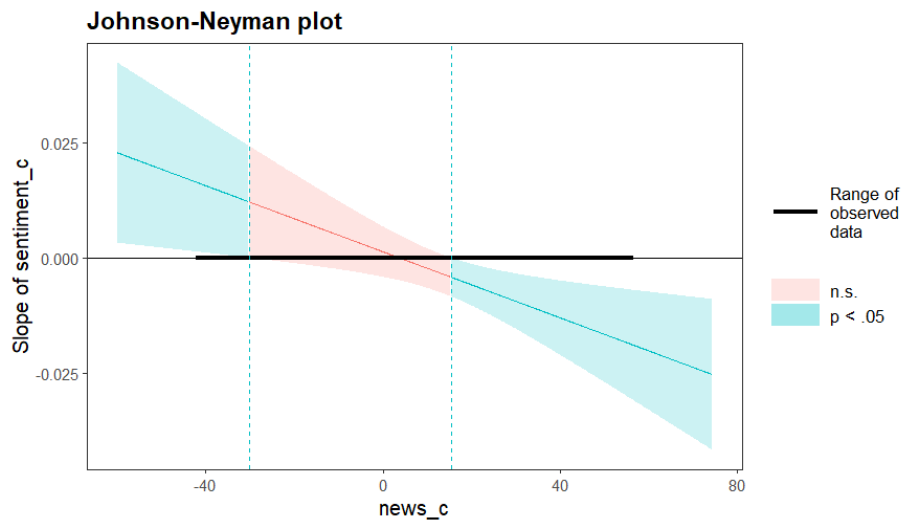


Figure 4: Johnson-Neyman plot for Parkinson volatility. The confidence interval for the simple slope of sentiment_c includes 0 when news_c lies within the range of $[-30.28, 15.44]$.

Garman-Klass model

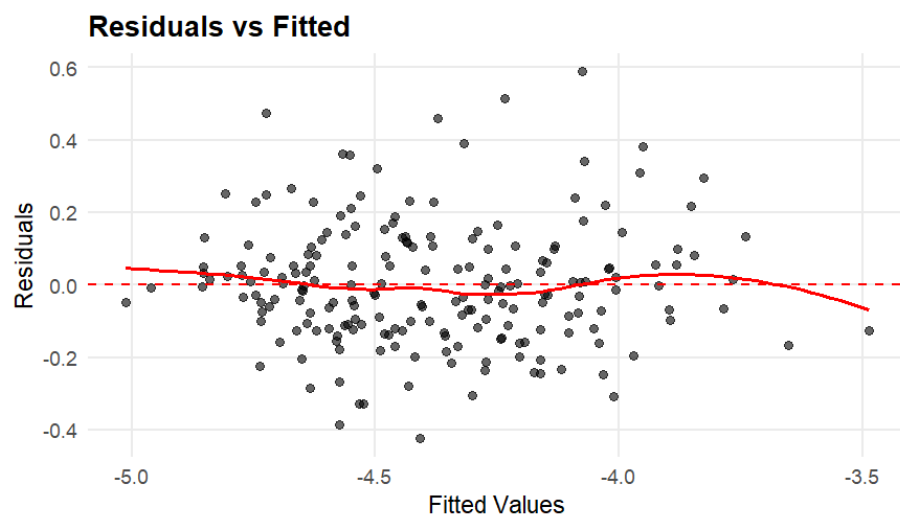


Figure 5: Residual plot for the Garman-Klass model.

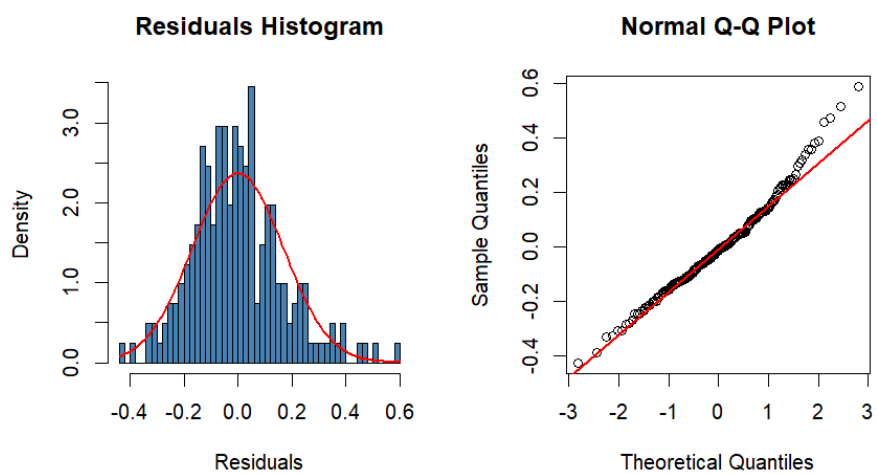


Figure 6: Histogram and QQ plot of residuals for the Garman-Klass model

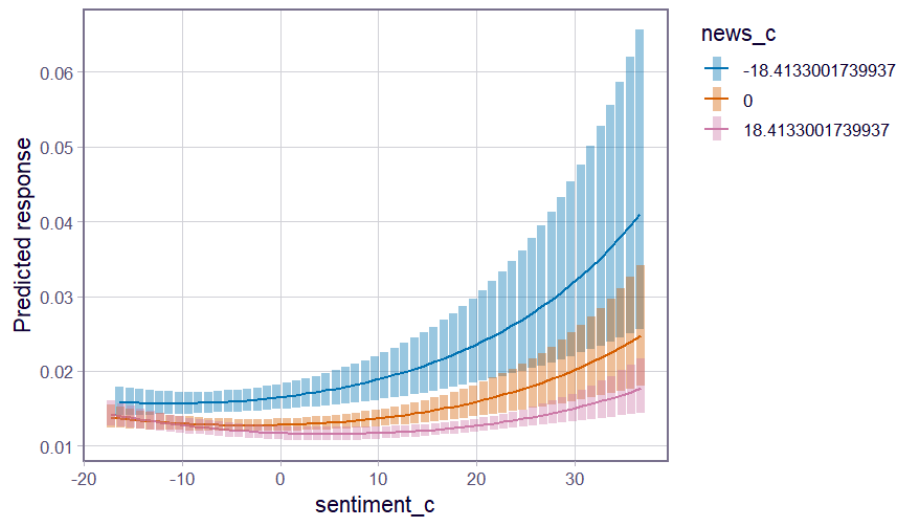


Figure 7: Emmip plot for Garman-Klass volatility for three levels of news (low, mean and high)

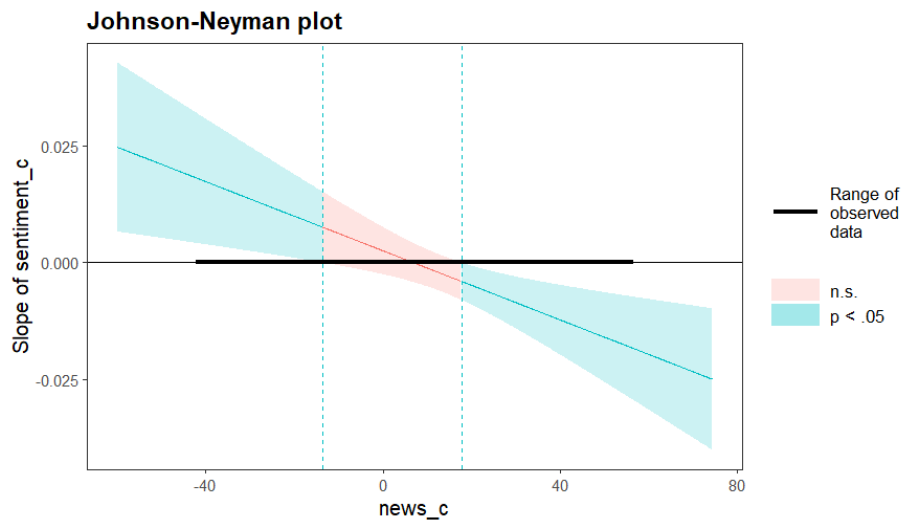


Figure 8: Johnson-Neyman plot for Garman-Klass volatility. The confidence interval for the simple slope of sentiment_c includes 0 when news_c lies within the range of $[-13.64, 17.77]$.

Rogers-Satchell model

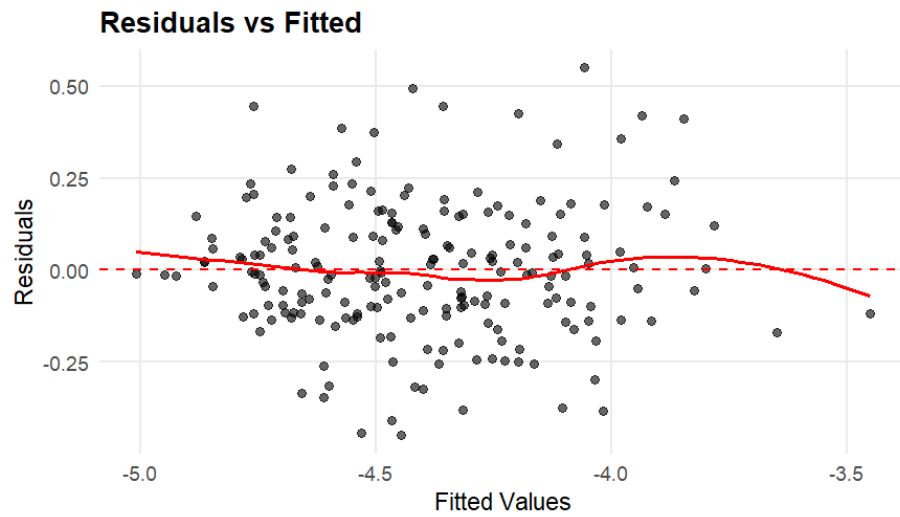


Figure 9: Residual plot for the Rogers-Satchell model.

Shapiro Wilks test yielded a p-value of 0.09935, indicating that the residuals are normally distributed.

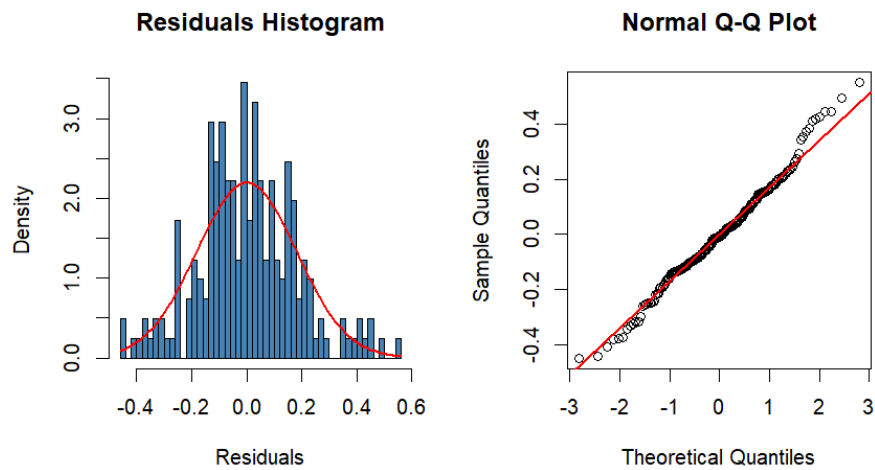


Figure 10: Histogram and QQ plot of residuals for the Rogers-Satchell model

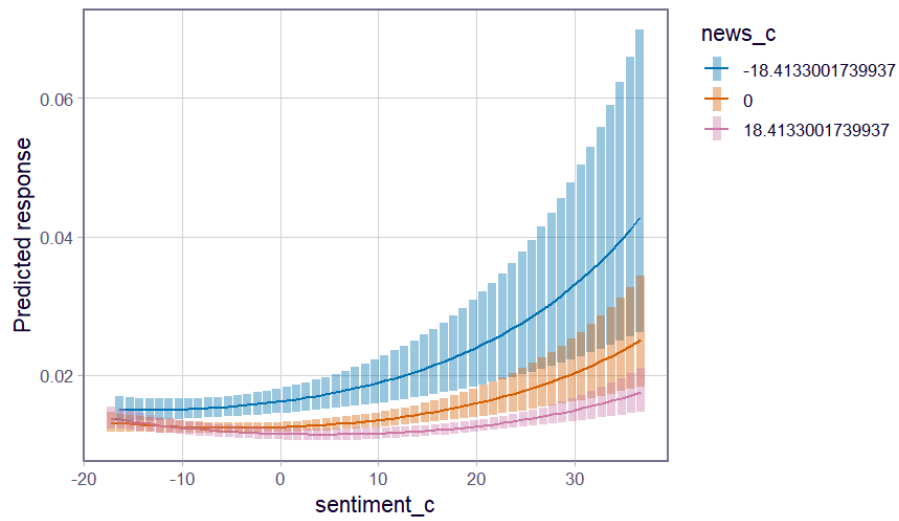


Figure 11: Emmip plot for Rogers-Satchell volatility for three levels of news (low, mean and high)

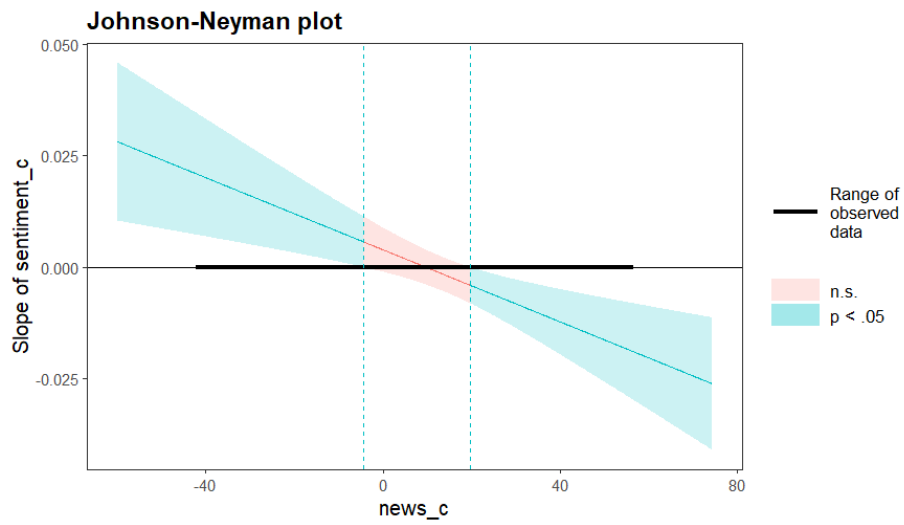


Figure 12: Johnson-Neyman plot for Rogers-Satchell volatility. The confidence interval for the simple slope of sentiment_c includes 0 when news_c lies within the range of $[-4.39, 19.76]$.

Yang-Zhang model

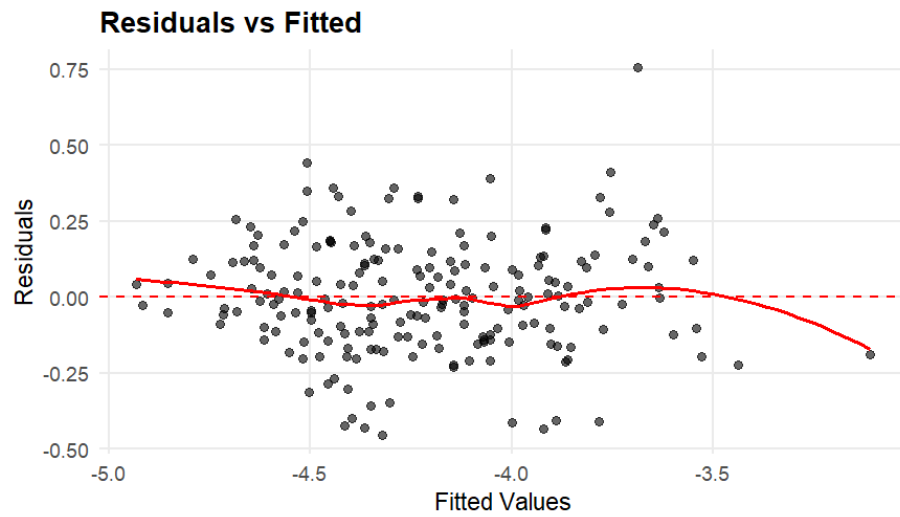


Figure 13: Residual plot for the Yang-Zhang model.

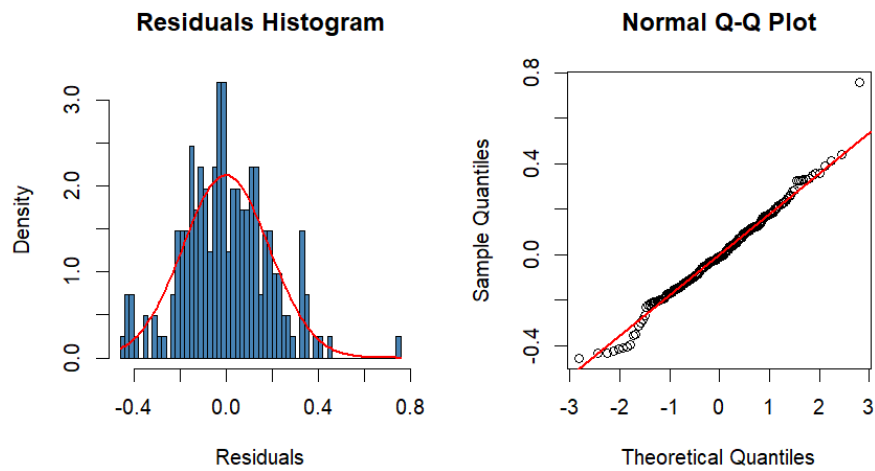


Figure 14: Histogram and QQ plot of residuals for the Yang-Zhang model

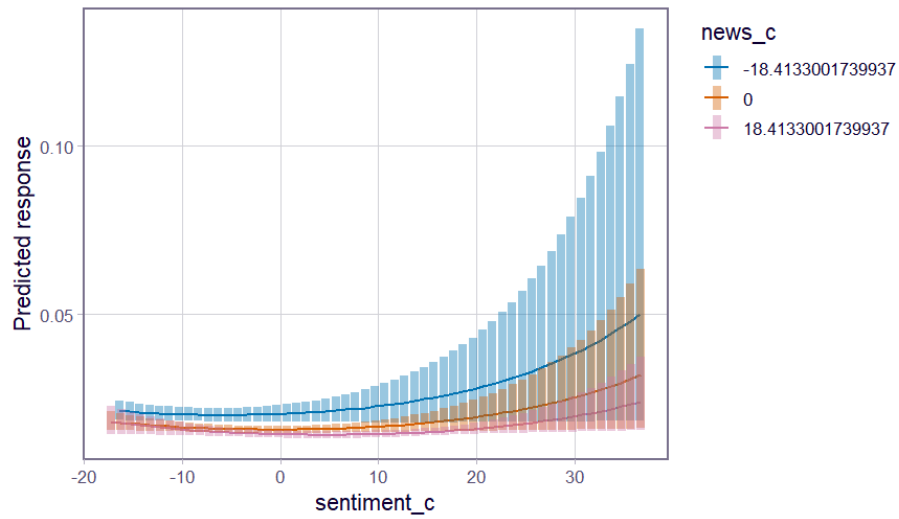


Figure 15: Emmip plot for Yang-Zhang volatility for three levels of news (low, mean and high)

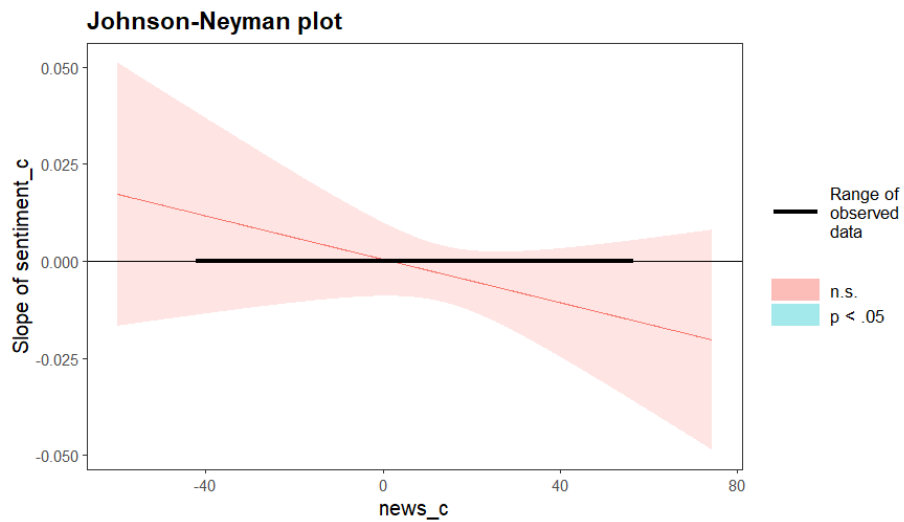


Figure 16: Johnson-Neyman plot for Yang-Zhang volatility. The confidence interval for the simple slope of sentiment_c includes 0 for the whole range of observed values.